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 Lab 3- My Shiny App - Punta Cana Airbnb ROI Simulator
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Link to My Shiny App
https://djgibbens.shinyapps.io/PuntaCana_ROI_Simulator/

Given the recent volatility in the stock market, I've decided to diversify my Roth IRA by exploring real estate opportunities in the Dominican Republic.

Recently, I came across the following investment opportunity
<https://www.jardinespuntacana.com/>

The developer provided the rental and profitability data noted in the screenshot below.

RETURN
1 BEDROOM

PRECIO	79.800 USD
ADM. RENTA 20%	256.02 USD
MANTENIMIENTO	84 USD
ELECTRICIDAD	120 USD
INTERNET	50 USD
R.O.I	11.58%

Ingresos	
Ingresos brutos	\$1,280.10
Ajustes	\$0.00
Tarifa por servicio del anfitrión	-\$37.65
Retención de impuestos	\$0.00
Total (USD)	\$1,242.45
Impuestos	
Retención de impuestos	\$0.00
Impuesto remitido	\$0.00
Impuesto liquidado por el anfitrión	\$0.00
Impuestos liquidados por Airbnb	\$0.00
Estadísticas de rendimiento	
Noches reservadas	24
Promedio de noches por estadía	2.7

Upon analyzing the data, in Excel, I noted that the net profitability per night was \$30.52 USD.

Ingresos Brutos (Revenues)	\$ 1,280.10
Ajustes	-
Tarifa por servicio de anfitrión	(37.65)
Retención de impuestos	-
Total (USD) - Gross Margin	1,242.45
Adm. Renta	(256.02)
Mantenimiento	(84.00)
Electricidad	(120.00)
Internet	(50.00)
Ingresos Netos Mensuales (Monthly Net Income)	\$ 732.43
Ingresos Netos Anuales(Annual Net Income)	\$ 8,789.16
Precio (Acquisition Cost)	\$79,800.00
ROI (Net Income/Acquisition Cost)	11%
Reserved Nights	\$ 24.00
Calculated Nightly Rate	\$ 53.34
Occupancy Rate	80%

Ingresos Netos Mensuales (Monthly Rate)	\$ 732.43
Reserved Nights	\$ 24.00
Nightly Rate (net)	\$ 30.52

Following in the footsteps of the Mexican saying, “te creo pero te verifico” (trust but verify), I downloaded the following data set from

<https://www.airroi.com/data-portal/markets/punta-cana-dominican-republic>

Available Datasets

Listings Data

Comprehensive property listings with detailed information including property attributes, ratings across multiple categories and various trailing twelve months performance metrics

CSV JSON Parquet

Download

which I saved as a csv to create the shiny app referenced here - https://djgibbens.shinyapps.io/PuntaCana_ROI_Simulator/

by David J. Gibbens

Choose a listing:

TTM \$26 - FULL FURNISHED ONE ROOM APARTMENT

Listing Summary

Listing: FULL FURNISHED ONE ROOM APARTMENT

Bedrooms: 1

Baths: 1

Room Type: entire_home

Overall Rating: 4.8 ★

Performance window:

☒ TTM (last 12 months)

☐ L90D (last 90 days)

☐ L90D (last 90 days)

Seasonality multiplier:

0.5 1 1.5

Base nightly rate (USD):

40 2,000

Occupancy rate (0-1):

0 0.8 1

Cleaning fee per stay (USD):

0 300

Number of simulations:

200 5,000

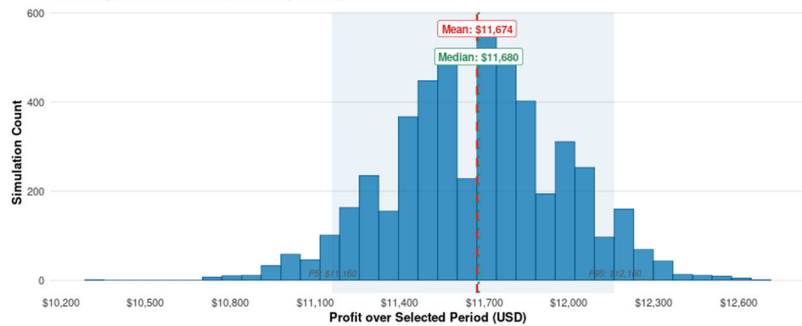
Number of days in period:

7 365

Simulated Profit Distribution

Simulated Profit Distribution

Shaded region = 90% of simulated outcomes (P5 to P95)



Summary Statistics

Summary Statistics

Profit over period (USD):

mean	median	sd	p5	p95
11674.41	11680.00	303.07	11160.00	12160.00

Profit per booked night (USD):

mean	median	sd	p5	p95
40	40	0	40	40

Profit per stay (USD):

mean	median	sd	p5	p95
160	160	0	160	160

Slider Guide & Assumptions

- Performance Window:** Toggle between TTM (Trailing Twelve Months) and L90D (Last 90 Days). This sets which historical period the pre-populated values are pulled from.
- Seasonality Multiplier:** Adjusts the nightly rate up or down to reflect high season (>1.0) or low season (<1.0). A value of 1.0 means no seasonal adjustment.
- Base Nightly Rate (USD):** The average nightly price for this listing. Auto-populated from the CSV dataset. Typical Punta Cana listings range from \$150–\$250/night.
- Occupancy Rate (0-1):** The probability a given night is booked. Auto-populated from the dataset. A realistic range is 0.55–0.70 (55–70%).
- Cleaning Fee per Stay (USD):** The cleaning cost deducted per stay. Auto-populated from the dataset. Most listings charge \$40–\$80 per cleaning. The simulator assumes 1 cleaning per 4-night stay.
- Number of Simulations:** How many random scenarios to generate. Higher values (1,000–5,000) produce smoother, more reliable distributions.
- Number of Days in Period:** The time horizon for the simulation. Use 30 for a monthly snapshot. Increase to 90 or 365 for a smoother bell-shaped curve — this is the Central Limit Theorem in action.

As luck would have it the very first listing “TTM \$26 - FULL FURNISHED ONE ROOM APARTMENT” is comparable to the proposed Jardines investment.

Although my shiny app pre-populates the parameters to the data in the data set, I made the following adjustments:

Slider Guide & Assumptions

- **Performance Window:** Toggle between TTM (Trailing Twelve Months) and L90D (Last 90 Days). This sets which historical period the pre-populated values are pulled from.
For my analysis, I used the trailing twelve months, to account for seasonality, given that Punta Cana has high season (December through April), low season (June to November) and two shoulder months (May and November)
- **Seasonality Multiplier:** Adjusts the nightly rate up or down to reflect high season (>1.0) or low season (<1.0). A value of 1.0 means no seasonal adjustment.
I used a value of one given that I conducted a full year analysis (365 days) and trailing twelve months of revenue.
- **Base Nightly Rate (USD):** The average nightly price for this listing. Auto-populated from the CSV dataset. Typical Punta Cana listings range from \$150–\$250/night.
I used \$40.00 per night which was the closest value on the slider to the \$30.52 of net revenue calculated on the prior page.
- **Occupancy Rate (0–1):** The probability a given night is booked. Auto-populated from the dataset. A realistic range is 0.55–0.70 (55–70%).
I used the occupancy rate of 80% (24 nights reserved out of 30.)
- **Cleaning Fee per Stay (USD):** The cleaning cost deducted per stay. Auto-populated from the dataset. Most listings charge \$40–\$80 per cleaning. The simulator assumes 1 cleaning per 4-night stay.
I used a value of \$0.00 given that the cleaning charge is reflected in the net base nightly rate.
- **Number of Simulations:** How many random scenarios to generate. Higher values (1,000–5,000) produce smoother, more reliable distributions.
I used 5000 simulations to approximate a normal distribution.
- **Number of Days in Period:** The time horizon for the simulation. Use 30 for a monthly snapshot. Increase to 90 or 365 for a smoother bell-shaped curve — this is the Central Limit Theorem in action.
I used 365 days to get a smoother distribution

Preliminary conclusion – more analysis is needed. Although the developer's ROI screenshot indicates that the occupancy rate is 24 nights out of 30, the highest occupancy rate for any property in the Air Roi data set was 49.6%. Assuming an 80% occupancy rate (24 booked nights out of 30), the profitability in the shiny app approximates the profitability noted in the developer's ROI analysis.